Assignment

Artifical Intelligence

Qno1:

(i)

objective:

To apply Minimax and Alph-Beta pruning on a Game tree. complete the optimal root value and more, and show that alpha-beta prving reduce the number of evaluated leaves while Giving exactly the same Minimax Decision.

(ii)

Case stuoy Input Data:

Two players play a zero-sum game. Max tries to get the highest score. MN tries to get the lowest case. The tree is a complete binary tree of depth 3 (8 leaves) The leaf utilities were Generated randomly (random.seed(6)) integer From 1 to 20) and then kept Fixed, so the code graph and this all use same values.

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Leaf value (left to right):

[19, 3, 9, 2, 1, 5, 19]

Node name: A = root (max); B, C = Min node; D, E, F, G = MAX node leaves 1 to 8

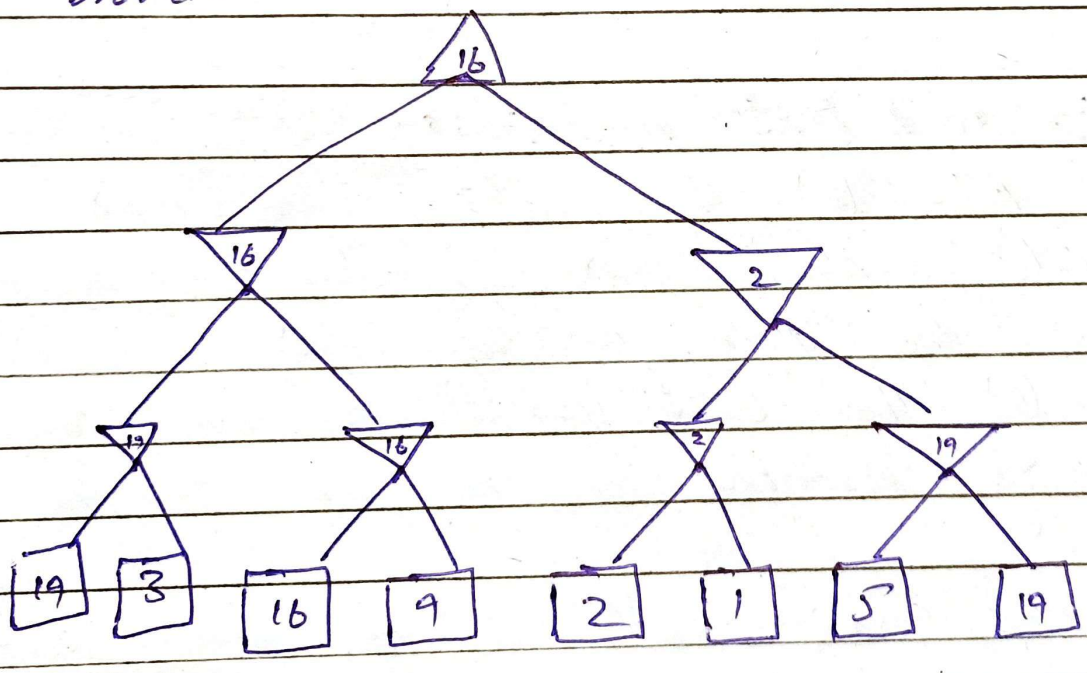
Structure:

A -> (B, C); B -> (D, E) C -> (F, G)

D -> (19, 3) E -> (16, 9) G -> (5, 19)

Legend:

orange tringle (up) = MAX tried to Get the highest score, MIN tried to Get the lower score. The tree is a complete bottom MAX= node, green tringle (down) = MIN node, blue square = leaf, grey = pruned (never visited) line with x = pruneted branch.



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Step	Node	calculation	Result
1	D(MAX)	max (19, 3)	19
2	E(MAX)	max (16,9)	16
3	F(MAX)	max (2,1)	2
4	G(MAX)	max (5, 19)	19
5	B(MIN)	min (D, E) = min(19,16)	16
6	C(MIN)	min(F, G) = min(2,19)	2
7	A (MAX, root)	max (B, C) = max (16,2)	16

Task 2:

Leaf - Evalution counters:

A Global counter was added in the terminal (base) case of both function: minmix leaves For minmix and alphabeta leaves For alpha beta Each time a leaf score is read, the counter increase by 1. The Function keep the same Format as the lab manual.

Alfa and beta meining:-

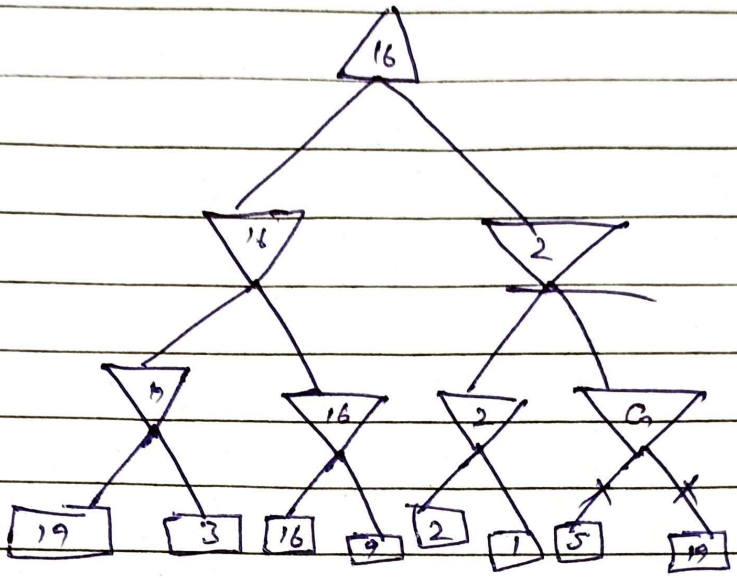
Alfa (α) the best value MAX (highest) value MAX can already guarantee on the path to the root.

Beta (β) The best (lowest) value MIN can already gurantee on the path to the root.

Pruning condition: if $\alpha \geq \beta$ the remaining children

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of that node cannot change the final decision, so they are skipped.



Node	Type	α (entry)	β (entry)	what happen
A	MAX	-inf	+inf	Start visit B first
B	MIN	-inf	+inf	visit D first
D	MAX	-inf	+inf	leaves 19,3 evaluated D = 19 B now has 19
E	MAX	-inf	19	leave 16,9 evaluated E=16 (<19, no prune) B=min(19,16)
C	MIN	16	+inf	visit F
F	MAX	16	+inf	leaves 2,1 evaluated
C	MIN	16	2	α=16 ≥ β=2 so, PRUNE C (leave 5 and 19 are never
A	MAX	16	+inf	A = max(16,2) = 16

Program output	Root value	leaves evaluated
minmix	16	8
Alfa-Beta	16	6

Saved by pruning somevale 2 leave (2 & 5)

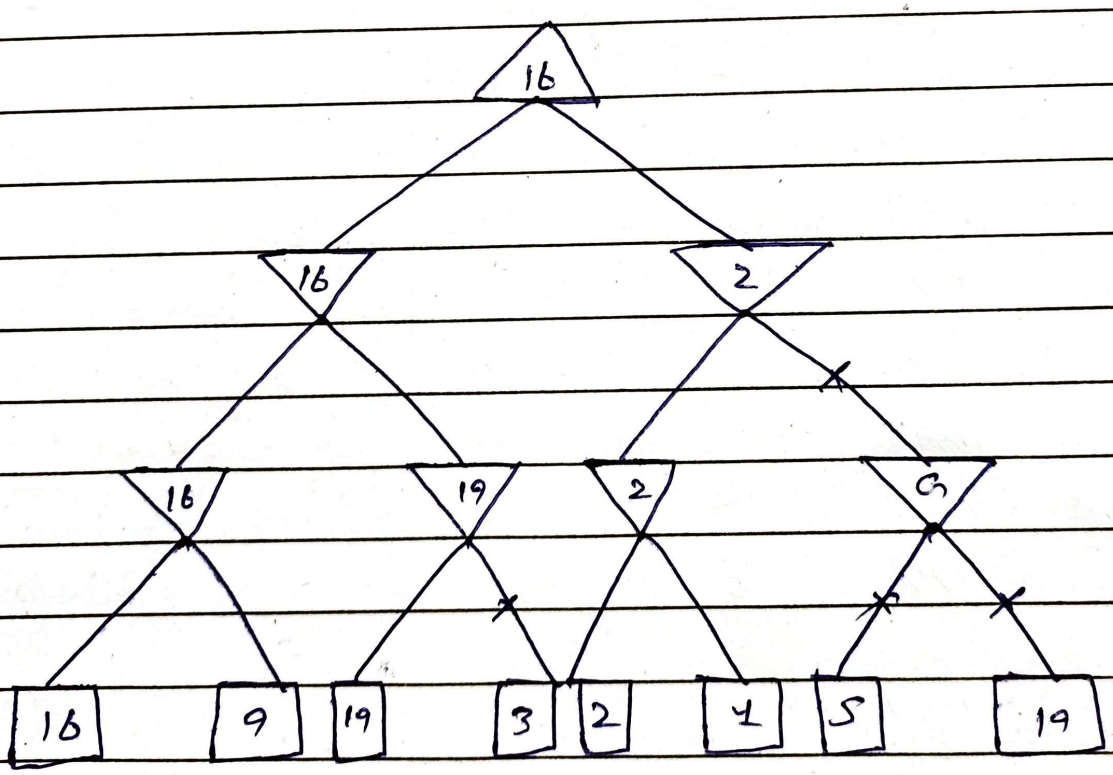
Both algorithm return 16 and root action is left (A to B), where left MIN=16

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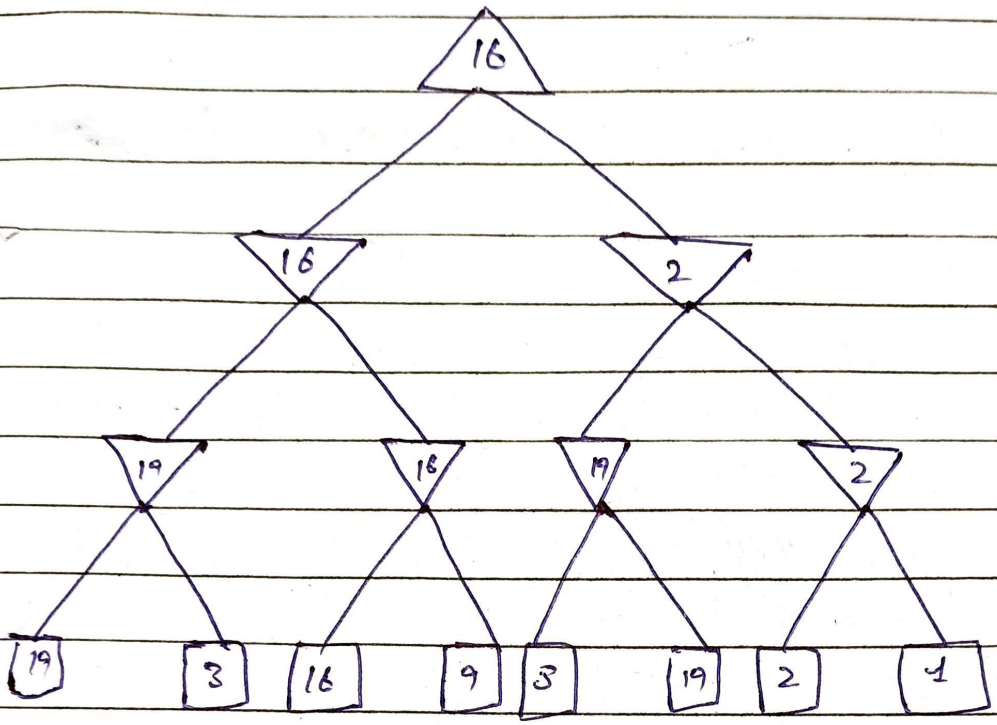
Task : 3

Reording Branches:

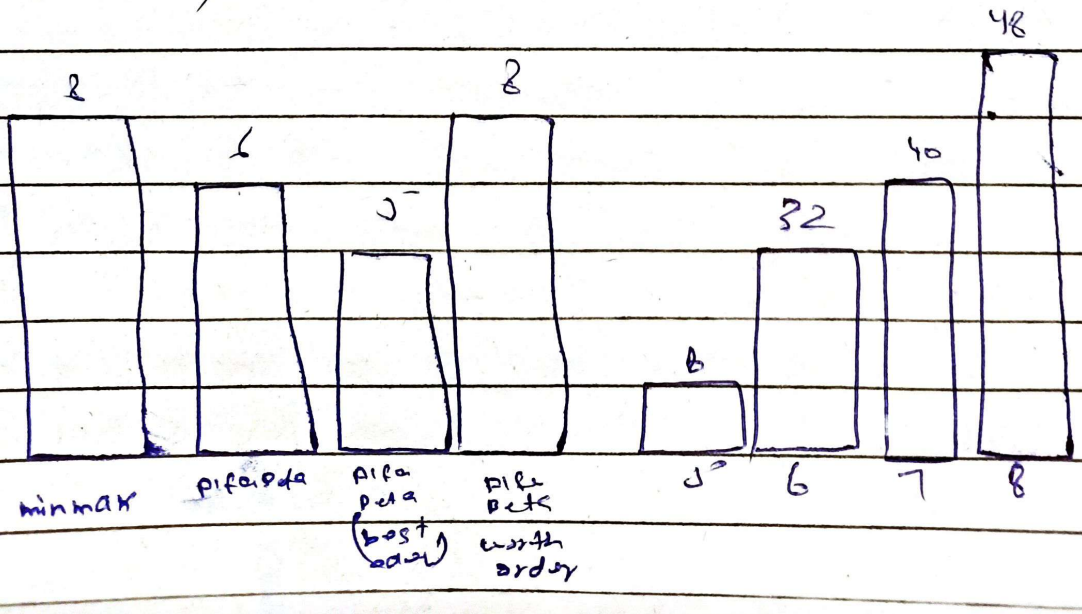
Reording Branches swapping the left and right children of internal node. The tree stay the same only the oder in which children are visited changes. The tree has 7 internal node (A to G) so there are $2^7 = 128$ possible orderings. All 128 were tested the program.



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Ordering	Leaves (left to right)	MM value	AB value	AB leave
original	[19, 3, 16, 9, 2, 1, 5, 19]	16	16	6
Best	[16, 9, 19, 3, 2, 1, 5, 19]	16	16	5
Worst	[19, 3, 16, 9, 5, 19, 2, 1]	16	16	8
Reversed (all swapped)	[19, 5, 1, 2, 9, 16, 3, 19]	16	16	8



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Task6

Edge Cases and comparison:

All value	Leaves	Value	MM leave	AB Leave
All equal	[7, 7, 7, 7, 7, 7, 7, 7]	7	8	
All negative	[-5, -12, -3, 20, -8, -1, -9, 15]	-5	8	
Smart tree	[4, 11]	-11	2	
16 random leaves	[11, 27, 45, 41, 19, 31, 10, 31, 33, 12, 83, 34, 16, 1, 1]	19	16	

Minmax and alfa-beta gave the same value every time, and alpha-beta never used more leaves.

Average leases: minmax = 16.0 AlfaBeta = 12.0 (25% fe

Intresting case: with all equal leaves pruning still happen (5 to 8 leaves) because the condition is $\alpha \geq \beta$ which also trigger when the two bounds are equal.

Limited / Faire analysis:

The manual's minimax drive pads the scores list with 0 when the number of leaves is not power 2 leaves Those fake 0 leaves are treated like real ability and can change the result